

ActInf Livestream #024.1 ~ "An empirical evaluation of active inference in m

Livestream | 1:50:42

hello everyone

thanks for joining and welcome to actin

flab

live stream number 24.1 today it's june
22 2021 and we're going to be discussing

the paper and empirical evaluation of
active inference in multi-armed bandits

with several of the authors so thanks

everyone for joining

welcome to the active inference lab we

are a participatory

online lab that is communicating

learning and practicing applied active

inference

you can find us at the links here on

this screen

this is recorded in an archived live

stream so please provide us with

feedback

so that we can improve on our work all

backgrounds and perspectives

are welcome here and we'll be following

good video etiquette for live streams

this short link has a schedule of all

the live streams that we've been doing

and will do for 2021

and we're here today on june 22nd in

number 24.1

which is the middle of the three-part

series on this paper about multi-armed

bandits

the zero video 24.0 had

some context and some background on this

paper that we're going to be

discussing an empirical evaluation of

active inference in multi-armed
bandits and today we're here with
three of the authors so thanks so much
for all of you who are joining today
because we'll have a lot to discuss and
learn about and
in today's discussion for 24.1
we're gonna first just go for some
introductions and then we're going to
have
a presentation by the author and then
we're going to just open it up for
discussion so if you're
here on the video call or if you're
watching
live in the live chat just feel free to
ask any question
and we can go wherever people are
interested
in going so that being said here we are
in the introductions
we'll just go around and introduce
ourselves and
say hello and then especially for the
authors who are joining for the first
time
it'd be awesome to hear anything you
want to
say or we'll ask you questions as well
so
i'm daniel i'm a postdoctoral researcher
in california and i'll pass to
dave i'm
in the tropical mountainous rainforest
120 miles north of manila
my background is in cybernetic learning
theory
general psychology and machine

translation

but not much math so i'm floundering

with much of the active inference

world we'll go to sarah

and then continue on

i'm sarah i'm a postdoc in treeson

together with tim t

my original background is physics and

biophysics

but i wrote my dissertation essentially

about active inference

and especially applied to habit learning

and yeah that's it for me

and going on to dimitri

should we let her introduce himself i

don't know i'm not sure if he wants to

do that

okay i mean i can also introduce him

shortly so

there was our colleague from ucl

previously so he worked in max planck

ucl center for computational psychiatry

and aging research and currently he has

switched to industry so he is working in

second mind

uh on the applied reinforcement learning

there so

basically he is a expert on this other

part of staffordshire

which doesn't cover active inference

right

so and uh yeah i'm also the postdoc

technical university of dresden uh both

me and sarah are at the chair for neuro

imaging

uh where like the head of the chair is

stefan kiebel

uh and and uh yeah we have been involved

with active inference for a while now
since probably 2015-16
and have been applying it to various
questions regarding human behavior
cognitive control decision making in
dynamic environments and similar
um and yes so um
should they kind of switch to the slides
now or
sure we can go to the students or if i
can just ask one
general question to any of the authors
who wanted to respond
was it that you were interested in
active inference and then
looking for domains to apply it in
or were you interested in a domain and
then sort of found
active inference as a way to integrate
what you were working on
i yeah well i mean so my background is
also like right in physics
complex systems uh and computational
neuroscience and this physicist we like
to
think about this unifying theories so
in this sense active inference has this
appeal that it can collect
connect a very kind of distinct way of
basically thinking about decision making
thermodynamics um right uh stochastics
dynamics and
uh very kind of different areas of
of understanding dynamical systems so to
say
uh and kind of in control so that
that would be the appeal and uh right uh
i mean definitely uh as a tool to apply

like to decision making human
understanding human behavior
this is kind of where this all started
right
so what new can we learn basically
uh using this approach
and sarah any thoughts about especially
i'm curious about the biology side
because we hear a lot about the
math and the physics verging towards
active inference but it's also cool to
hear about biology and that was
my background as well my background is
rather in biophysics
so i'm only tangentially in biology
but what i like is also that it connects
with some general information processing
scheme in the brain
and actually my masters see this i was
working way way down
in the abstraction hierarchy on spiking
neural networks and like receptor
dynamics
and actually in this area i find it easy
to not see the forest for the trees
and i think actually connecting upwards
and asks the question what is a general
information processing that's going on
and then connecting it back downwards
again is what also
attracted me to active influence
awesome so as usual people will be uh
i guess joining or leaving i'll unshare
this
screen and dimitri if you want to uh
jump into the presentation that'd be
awesome
yes

okay so you see yeah
and uh all crop everything so go for it
excellent okay so
let me start with a bit kind of
motivation for this work uh
so as well you realize kind of our
involvement with active inference
came from this kind of cognitive
neuroscience
uh direction uh
and originally not so interested in the
technicals or machine learning side of
it
uh however if one thinks about uh
multi-arm problem bandits right that's a
very general uh
problem which kind of generalizes
resource allocation problems
then when they realize that this is most
of kind of behavioral
experiments can be cast into this kind
of framework right
and once these this in the range of kind
of uh
extreme experimental cognitive
neuroscience uh
domains like decision-making in dynamic
environments
value-based decision-making structure
learning and similar uh
one can also think about like um
attention
as a resource allocation problem right
so it's
kind of there are many other domains
where
maybe they're not explicitly talking
about multi-arm bandits

as such but they can also be cast uh
into this uh
general framework and for example for
from decision making dynamic
environments one of the most
well-known kind of tasks is a
probably like reversal learning task
right that's kind of used in uh
hundreds of papers and this is also kind
of
what initially motivated me to make this
paper
uh it's a kind of just uh understanding
so if i apply active inference to liver
reversal learning tasks how does this
compare to other
like alternative decision making uh
approaches we can apply to that right
however right besides this kind of
cognitive neuroscience uh
uh direction uh multiple bandits have
like a range of industrial applications
uh they really use like
you know starting from marketing to
finance like recommendation systems in
finance for trading applications
uh i even in many like optimization
problems in the
like deep learning right you know there
are lots of
papers actually showing how you can
speed up learning by
um uh finding kind of better sets of
examples for
uh for the deep learning uh uh
systems uh and right so basically the
way this also kind of
allows them active inference to bridge

this into a machine learning gap and
find
new uh applications there if it's fine
if it kind of shows to be useful for
this kind of multi-armed setup
as kind of adding something new to the
existing work
right and
when we talk kind of multi-arm bandits
have a
range of different formulations
today we will talk about stationary
bandits and dynamic bandits
uh where just means that uh reward
probabilities either are fixed over time
or change
uh in different ways however people also
write uh
discuss adversarial bandits risk of air
bandits uh contextual bandits
one can also talk about non-markovian
bandits or that
kind of rewards uh depend on a sequence
of actions or
have kind of some memory dependence in
the system
and so on right they're really like a
range of
different uh kind of structural
definitions of multi-arm bandits which
kind of then require a different way of
thinking about the problem
uh and this is potentially like
something uh also interesting for the
future
expanding uh what we did here uh into
this
other domains and seeing if again this

the results are generalizable to other

uh

definitions of multi-unbending problems

uh okay so just

um how i structure the slide so far

we have like two parts one isn't about

stationary bandits

and then we will switch which will go to

switching bandits

yeah uh and uh yeah given that we have

two hours i hope that will be enough

time but

if yeah if not we can also continue for

the next session

for example with switching bandits or so

let's see let's see how this goes

um okay so what is stationary bandit so

the

definition of the problem is as follow

right on any trial

an agent makes a choice between k -pops

uh choice outcomes are binary variables

so we kind of focus here on bernoulli

bandits

outcomes can also be drawn from any

other distribution so

and then for example you talk about

gaussian bandits or

uh yeah depending on the under how the

rewards are

generated we will just uh work with

bernoulli here

here bandits and we have

we have fixed uh reward probabilities in

a very specific way

so there is only one arm which has a

reward probability associated with $p_{\max}$

which is one half

plus some uh term epsilon like which is larger than zero and all other uh um all reward probabilities associated with other arms are just fixed to one half

but this kind of allows us to like control the difficulty of the task so because the more smaller the epsilon is the closer to zero is uh the more difficult is to distinguish the best arm from from others and the more samples one he needs to draw to kind of realize the difference right

uh and right beside this like the best arm advantage term uh one also realizes that right number of arms k is also increasing the difficulty the more arms you have the more time you need to figure out what is the correct arm to play okay so just to give you an illustration of the example right we have forearm banded here uh so whenever kind of agent pulls one of the arms uh it gets it gets either zero or one so we can think of this as a reward or absence of a reward right uh beliefs about the reward probabilities uh we will assume that right these are beta distributed so we will use beta distribution to model kind of a representation of reward probabilities associated with each arm and we will use um bayesian belief

updating for
all action selection algorithms so this
also kind of limits
uh the range of algorithms which we want
to compare
here in this study
i will explain a bit uh later
why i mean we focus on that
uh so
just to give you a kind of visual
example of what this does
is right uh whenever an arm kind of
an agent starts with with kind of flat
beliefs over
reward probabilities associated with
each
arm so this is like a uniform
distribution
uh this is a special case of beta
distribution
and whenever uh it pulls like selects
one arm
uh it gets uh either one or zero as an
outcome
and it updates right uh the beliefs
about the reward probabilities uh in
one of the directions right uh so
in here right we're kind of it's
sampling one more often so it
has increased belief that there is a
higher reward probability
associated associated with this
uh so formally how
this is kind of implemented what you see
here
is like to a very simple you know
generative model uh
so with each k arm we assume there is

some associated
unknown reward probability that okay
which is just a number between zero and
one
and uh choice outcomes of choices are
either zero or one so they're just
binary choices
so this kind of constraints are like uh
observation likely
to a bernoulli distribution so this is
just kind of
product of different bernoulli
distribution depending on which arm one
is selecting
and this kind of uh term below like this
is
our prior belief about reward
probabilities
and this will assume is just kind of
product of uh many uh
beta distributions where here alpha zero
and beta0 are just fixed to one so this
corresponds to the uniform
right right one can also start with
other values but uh
i think this makes some sense again
reflects uh no knowledge a priori about
report
uh so now given some choice at
trial t_0 uh we just apply
by subject rule and we get from some
prior beliefs
uh a posterior beliefs and then i think
about the stationary case is if
our priors are beta distributed our
posteriors will be also
beta distributed and the update
one only needs to take care about how we

update write the
parameters of beta distribution alpha
and beta
and this just works in a form of
accounting basically
alpha counts how many times you observe
one as an outcome and beta counts how
many times you
observe zero
and so because of all this uh in this
stationary case
inference is uh exact so that because
both prior and possible belong to the
same distribution family you have kind
of conjugate prior
here setup so you can just track
effectively
the updating there is another kind of
uh representation of the update rules
one can use here so for example
if we express the updating in terms of
expected kind of outcome μ
and the scale parameter ν here we see
that actually the expectations are
updated in the form of a
delta like learning rate so right where
this kind of learning rate is something
which decreases over time so basically
new is just increasing over time
whenever you select specific
arm and you will just increase by one
and this means that the learning rate
decreases over time so
the more you sample from one arm the
less you will update
your beliefs about it all right so this
is uh
in practically what's happening here

very very simple um

okay so and uh basically this is

everything we need to know about kind of
this uh

let's call it perceptual part of the

generality model so how we update beliefs
given some

outcomes now we have to kind of
introduce how we select actions right
based on these beliefs and for this
right i

we kind of can talk about action

selection algorithms

and so uh

uh they are in kind of in the literature

there are lots of red examples one of
the oldest one is probably epsilon
greedy

you also mention it during the first

[Music]

presentation point zero tyler people

also talk about right uh ucb

uh upper confidence bound this is also
one of the oldest one

kl ucb uh it's kind of extension which
uses uh

kl as uh exploration okay kl divergence
is a

estimate of exploration uh bound

there is thompson sampling etc uh here
we will

focus mostly on on this

tree right we will use the optimistic

thompson sampling

as a one kind of comparative example and

another is bayesian upper confidence
bound

so first both of these algorithms have

been extensively
analyzed in the literature and people
just find something
they work in in many examples lots a lot
much better than kind of let's say
non-bayesian approaches
uh and secondly right we can
use basically for all three algorithms
the same uh
update rules because they are just
bayesian uh
they're just algorithms corresponding to
bayesian and bandits so
we can kind of our motivation here is to
compare kind of
action selection principles based on the
uh
action selection algorithm and not based
on potentially different ways how you
update beliefs about
uh about the history of observations
right
and uh so right this would be kind of uh
motivation and also a bit to simplify uh
the comparison i mean you can like add
to the list at least 10 other ways of
uh doing the rate action selection and
even like
decision making in multi-armed bandits
uh okay so just a bit too
kind of historic example important this
is just like
upper confidence bound although i will
not directly compare it it has some
relevance to understand also what is
happening kind of in
active inference later uh
and this uh this form here uh just

corresponds to
a version uh which is adapted to
specifically bernoulli bandit so some
people might be more familiar with
other version which was derived for
gaussian
bandits right so this is kind of in this
paper here
one can see uh these different
derivations and examples
how how this works what is important
right to notice here is basically that
so this first term is just the expected
probability of reward on arm k
and the second two terms correspond to
this exploration bonus or the bound
uh and uh
so this is uh typical for ucb is that
the
bound kind of increases over time
logarithmic
logarithmically with the number of kind
of trials you're doing so basically
the more you're not selecting one arm
the more you're kind of pushed
again to select it at some point in time
in the future
all right so basically this necessarily
increases over time
uh if you are not interacting with the
specific r
right uh but still although like where
the algorithm is very simple it has nice
theoretical
results so because it's kind of it's
efficient algorithm
it converges in infinite number of
samples

however right here we'll talk on this kind of asian variant of the upper confidence bound which actually make a select arms based on kind of the percentile uh upper percentage of cumulative distribution function right so so as time uh progresses uh you uh one is trying to estimate the uh like the extreme value of a believe distribution which in this case is just like a beta distribution and and one one gets this extreme by solving this equation which just corresponds to inverse regularized incomplete beta function so basically the more extreme point your beliefs contain the more likely is that i mean the more likely that you will select that arm so uh the more relevant is to select this arm because you're expecting that right uh this high value is still possible in the uh as a reward probability in this uh in this setup and so but this kind of algorithm has couple of parameters but what authors kind of show in this paper is if you have a very good results for just fix by fixing c to zero so basically this term just becomes one uh and so we we just

use their advice here in the paper so we
will not uh
and we didn't kind of try to analyze
other possibilities
or other kind of values of these
parameters in in the
uh in this case
uh so for um thompson sampling and this
is again one of the
classical algorithms first probably um
attempt with bayesian bandits so with
bayesian decision making
and it's also extremely simple right
given some beliefs
about probabilities at each arm you you
sample one point
uh so right if there are like 10 arms
you would sample from each arm one point
and
you will just select an arm which gives
you the
high largest values right uh
a variant uh of that which was shown
kind of last five to ten years that it
works slightly better is this optimistic
stomps on sampling
where uh one is constraining the samples
only to
uh values which are larger than the
expectation right so
so one is first making a sample from
each arm and then
if the sample is larger than the
expected probability
uh of reward then uh one keeps the
sample otherwise uh you just use the
expected value as uh
uh as a value a kind of a reward

probability associated with that right
and you're just again over different
arms maximizing taking the uh selecting
the arm which
gives you the maximum reward probability
but this is a kind of a very stochastic
approach
where this exploration actually kind of
bonus to this algorithm comes from
basically this random sampling from a
probability distribution and right the
broader your beliefs are about something
the more likely that you will get the
kind of large value
hence the more likely that you will
explore more
uh select that that arm uh
different runs and basically exploration
is kind of
here completely driven by uh the
noisiness of the sampling process itself
um okay so now we come to
basically active inference uh
version of this
so um we are kind of simplifying
here things from what people maybe know
uh
how active infrared is used normally so
first we will use like
rolling behavioral policies which means
that
agent is not like tracking history of
actions that perform
it just kind of a repeating step of
policies on every trial
and in this case the behavioral policy
just
corresponds to a single choice so

in this type of bandit problems we
we are kind of analyzing here like agent
cannot change anything in the
environment
hence planning is completely irrelevant
uh in a way you cannot kind of position
yourself in us
in space better over time so that you
kind of
need to plan something which means that
right
just kind of single choice policy
evaluation a single time step in the
future is sufficient to make uh
good decisions uh
and uh right uh uh generally expected
free
we will base our action selection on
expected free energy here
uh where this would be uh
a form which uh decouples into a risk
and ambiguity term but we can also think
about this problem as freight uh
estimating uh expected value of
of different arms plus the expected
information gain so how much information
we can extract from different arms if we
select
so
now let's assume that we know how to
compute expected free energy i will uh
go through details on next slides
uh normally like right posterior oval
policy
is estimated as a kind of mixture
between
expected free energy so kind of a future
uh expectation about what behavior will

do
plus the kind of the second part which
is the free energy about the past
kind of outcomes however because we have
like rolling policies
this term is just constant uh
for each policy is the same it has the
same value basically so the posterior
policy or where action just corresponds
then to the
to the soft max right over the expected
free energy
uh and normally one one things like
uh about choice selection this is
assembly
samples from the posterior however we
are here only interested in kind of
optimal choices so for us
basically gamma is just infinite value
and we are just selecting the
uh making sure choices about actions
which minimize expected free energy
so this this is useful parameter to have
if you need to kind of uh
fit a model to the behavior but for this
kind of a
practical application there is not much
gain
in adding another source of noise here
um okay so how do we compute the
expected free energy right
well that's in a way quite simple so we
just
have a couple of terms here one is uh
posterior over
posterior beliefs over reward
probabilities right
given is this like q term this is just

the product of beta distributions
we have marginal likelihood so
probability of observing
o outcome given some action θ . so this
is just
marginalizing likelihood over uh our
current posterior beliefs
and another term is here just the prior
preference over
different outcomes right and this is
really easy to parameterize because we
we work with binary outcomes and we can
just define here a single parameter
 λ
uh so the higher the λ is the
higher the preference
is of the agent to observe once
com relative to zeros right
and this uh λ parameter also kind
of has a
role of regularizing the amount of
exploration an
agent does because the larger the λ
is the more um
uh the more focus is uh
is set on the exploitation of kind of
making selections based on the expected
value instead of expected information
uh and so this final term right
computing ambiguity is basically just uh
computing expectation over the entropy
of different outcomes
right so which also has relatively
simple form here
uh because of bernoulli likelihood
so uh without going into all the details
this is basically how the expected free
energy looks like so this first term

here
is correspond just to the expected
negative of the expected reward
so because we are minimizing expected
free energy this effectively means that
we are maximizing expected reward
and the second term is just a very
complex sets of
set of like equations which
gives you estimate of the expected
information gain right
and the in a way kind of motivation is
just because one cannot really
understand what is going on here right
we have kind of logarithms of
expected uh reward probabilities but
then we have a d gamma function of
parameters and
uh so instead uh we can kind of try to
simplify the term by
approximating right effectively the
information gain part
uh just to get better understanding of
uh
how basically expected free energy
scales uh
like with repeated choices and uh so
what
one ends up with very simple form right
basically the exploration term
just corresponds to 1 over
 2 divided by the number of time when
selected
different arms right so this is in a way
quite similar to
what what we had in the in ucb algorithm
right the like first exploration term
was also divided by something which is

proportional to how many times you
selected
uh an arm with but without write this
kind of
um x expansion of exploration bonus with
time so there is no logarithm of t
and i mean this will have a consequence
on
efficiency of active inference so i will
show you this in a moment
uh so how you wanna achieves basically
this simple form is just by
uh assuming that uh for large uh the
this the gamma function has this type of
approximation when x is
sufficiently large right so this just
means when you
sample sufficient number of time each
arm this will be
more like very accurate approximation
and we see that
this approximate so algorithm and the
exact they behave very similar so
it is in a way reasonably good
approximation
so you actually uh danielle i think you
asked like a question uh
about the scaling properties of this
right
one kind of uh
motivation for introducing approximation
is that in a way the algorithm is much
simpler so it scales better so you can
kind of run it
on much more arms easier
uh however like right the advantage is
not huge so maybe
uh it scales slightly better but you're

maybe gaining like 10
or 20 performance in computation time
uh so it's not something like uh which
destroys completely the exact form
at least not in this example right
because the problem itself is simple
however it helps us a bit understand but
just get intuition what is happening so
and uh at least like right how the
exploration bonus changes with time
uh okay so
before i kind of just show some results
uh uh
like comparison of different algorithms
i want to introduce
uh just concepts which come from uh
machine learning analysis of multi-arm
bandits uh which people use like
just to kind of rate the uh and compare
different algorithms right
how good they are in solving this
task
and this is done by using something
defining a regret
so basically uh regret is simply a
difference between
uh what you did at trial t minus what
was the best choice
that uh at that trial
and uh so kind of
uh assumption here is that there is some
oracle
which is solving the task which knows
exactly what was the best
choice in every trial right
and so normally people kind of consider
two uh two quantities either a
cumulative regret which is just

some uh of of regret
over or up from first trial till the
current one
or a regret rate which is just the
average over time of cumulative regret
right
and we will here use both just for
visualizing different aspects of the
algorithm uh and a kind of in stationary
case at least this is a very kind of
important result
because all for all good algorithms one
would expect that this regret
goes to zero over time right so as you
go to
infinite number of trials you should be
able to
always do like a good choice
uh and if this is the case then one can
show that uh
algorithms which have this property they
are called also sympathetically
efficient
and they scale for large t 's as
something
some terms times a logarithm over t
right
uh and so this is kind of um
uh one important aspect of uh
a slice for stationary case of different
decision making algorithms
so multi-arm bandit algorithms so they
should
at least scale as logarithm of t when
you expand
uh right when you go to large t limit
otherwise right this first thing will
not

probably hold well or they can also
scale slower than logarithmic but
for example if they increase linear
limit t

this will not hold anymore

uh okay so um

now let's go to the comparison right uh

we will look into how the missing

thompson sampling bayesian upper
confidence bound

exact different inference algorithm and
the approximate one

uh compared to each other uh

i will start first with just trying to

see what would be kind of a good value
for this λ parameter

uh in different settings uh

and also to give some kind of initial
comparison of

exact uh algorithm and the approximate
one

so what we are i'm showing here is a
regret rate right for different kind of
snapshots

so these dotted lines are after 100
trials

uh the dotted dashed line is after a
thousand trials

and uh the solid line will be after 10
000 trials right

and so what one kind of

notices here is that there is some
obviously kind of minima for different
 λ so this kind of

preference parameter uh

and that the longer kind of uh well
the more trials you do the smaller the
 λ should be so this this is not

quite nice and this kind of has a
consequence
however we can like just pick some value
which
uh so for example this purple dotted
line shows like
around 0.1 λ 0.1 which seems to
have
be close to minimum for most of these
cases right so we don't want to kind of
have different value for different
examples because i mean this is just not
practically feasible
you want to have a kind of general
algorithm which can be applied to many
different situations
at the same time so
what when we compare so bayesian ucb
domestic thompson sampling and just the
approximate active inference so
i'm just excluding this here the exact
one because
they will behave the same pretty much
for this specific parameter value
if we compare them in terms of
cumulative regret
we see that right the approximate active
inference is not asymptotically
efficient so this
curve just goes diverges over time
uh where's those if if you if you look
at the
kind of the great uh sorry the green and
the yellow curve they they kind of
flatten out
after some time and they get the slope
proportional to this dotted
lines which actually shows the slope of

this kind of uh
asymptotic limit right what you should
see for large t 's
now uh so why is this happening so the
thing is that
um because this exploration
bound is kind of just decreasing over
time
uh active inference algorithm kind of
gets stuck
into the wrong solution with some
probability which depends on the
difficulty of the task so right
the smaller the epsilon is uh the more
likely and
the more arms you have
sorry for smaller epsilon and for small
number
of arms there is a higher probability
that you get kind of stuck
right and one can see this if kind of we
take a snapshot
of of different runs so this is kind of
a distribution of cumulative regret
i'm just plotting the logarithm here so
like
making a histogram of over different
runs of this is like
10 000 runs of different algorithms
and just showing at what value for the
logarithm of the cumulative regret they
end up
as you can see here there is for kind of
active inference based uh algorithm you
see this kind of
well spike here and in the tail of the
distribution
which is proportional to just doing

random choices so that means that
basically algorithm was just selecting
wrong arm constantly
it never it never converged to a correct
solution

in a way it gets stuck into a rock
solution because the exploration bow
was reduced to too soon

uh right so in it

so this is kind of uh say limitation or
application of active inference to
stationary problems

because this is not the feature of an
algorithms you would like to have

in a way uh normally if

if you're kind of so example of this
would be

for example optimization problems that
you want to find

the best solution for a set of
parameters

uh now for example

by asian optimization finding the
minimum of what some unknown function
uses a thompson sampling this is kind of
seems to be a very efficient way of
finding the minimum however if you would
apply to such situation

an active inference basis um kind of
arm selection or sample selection uh
there is a chance that the algorithm
fails right you just kind of get stuck
in the wrong

minimum wrong optima it doesn't do
sufficient exploration

uh so this kind of uh requires

potentially some adjustments to how
actions are selected in at least in the

stationary case

uh so just to

okay

strange yeah i got some string slides

just to kind of

uh know a bit uh what the short term

behavior looks like

so from the perspective of kind of

cognitive neuroscience or like

human decision making you don't really

care about this asymptotic limit

because you don't usually expect people

to be in a either

stationary environment things always

change

or write that they kind of have to

repeat actions

so many times so what i'm showing here

now just

in a very reduced example so if we have

only three arms

and we just use kind of different

epsilon values so

uh task difficulties what is the

probability that

uh for different algorithms that you

select actually the optimal are

and as one can see that right initially

so by asian ucb for first maybe 25

trials

has the highest probability to select an

arm however there is a range of trials

like from

50 to maybe 1000 where

active inference based algorithm takes

over so right

in a way because of this information

game

term active inference is more efficient
in uh
targeting like the arms which will give
you the most information has
hence it can recover uh the best arm in
kind of some intermediate interval with
the highest probability
however as you kind of expand this after
like thousand trials or so you see that
this probability gets stuck so it never
converges to one
unlike the other ogre this is especially
evident for this difficult right
difficult problem small
epsilon and and so this is uh
in a way explanations of what happens so
basically algorithm although
it reaches good solutions after
it's hyper higher probability than other
algorithms
there are still a lot of uh um
lots of examples so in this kind of
simulations uh
parallel simulations which get stuck to
a wrong
solution and they cannot get out of this
so in a way right this kind of asked
question okay what can one do
to make active inference also synthotic
efficiency so how can the
maybe either generative model be changed
to
support increasing the exploration bound
over time
or maybe introduce kind of instead of
computing
expected expectations in um
of expected free energy one can also

just draw samples
from a posterior and kind of also
compute this
these two terms like information gain
and similar to like thompson sampling so
right there are kind of
different ways one can think of how to
add uh
exploration bonus another third option
would be to actually
introduce learning of the lambda
parameter
so that lambda itself kind of goes down
over time
so it kind of goes towards zero uh
in with specific uh
uh specific rules however uh yeah
currently we don't have very a good
solution for this
so we just leave it to this that has a
uh obvious limitation of just applying
active inference to this type of
problems um
are there kind of maybe any questions i
think uh
this would be like the half where we
switch now to the
perfect the other
um if anybody has any thoughts
definitely
i could ask some things or also
we'll ask if in the live chat people
want to post any questions
but um how much longer of a presentation
did you estimate that you had so that we
could kind of
also address some general points here
during this point one

um well i don't think there is more than
maybe 20 30
minutes max i didn't really
uh gauge it but there is less less
slides in the second question
would it make more sense to go through
it quickly
here yeah or would it make sense to do
it in the
next weeks uh i see which i think uh
both are fine for me i mean maybe
additional 15 minutes
okay so we'll complete the presentation
and in live chat
and on the video chat here we'll compile
our questions and then
in the remainder after your presentation
today and then next week we can have
more
open discussions so please continue
thanks okay
okay so uh now we
go to this like dynamic non-stationary
problem
uh and on any trial uh
uh in this case right it's this very
similar setup on any trial an agent
makes a choice between k
arms again we are focusing on the
bernoulli bandits so
outcomes are either just binary
variables
however what happens here is that the
reward probabilities associated with
each change over time right and we kind
of differentiate between switching
bandits uh where we will assume that the
changes happen at the same time on all

arms

and furthermore in switching bandits

they are kind of also called piecewise

stationary so there is like a period

where nothing changes and then there is

just one moment in time when

uh rewards reward probabilities on all

arms change

or uh we can think about a call of

another variant of dynamic bandits would

be restless bandits

where changes happen independently on

each arm and they are continuously

changing

over time so for example following a

random walk

i will only talk about switching bandits

but

from some testing i did

all the results generalized also to the

restless case

and in this kind of beside this

number of arms and the difference

between the best arm and the

other arms of kind of reward

probabilities epsilon and

k we have another task difficulty this

is the rate of change

or like just change probability so the

more often things change the more

difficult the task is first

especially if you have uh many arms

um and uh

i will consider like switching bandits

with fixed difficulty which

is just extrapolation of the stationary

case to

uh by introducing changes to about

to which arm is associated with the
maximal reward right
we will always have the same rewards on
on all arms it is just that from time to
time
optimal arm changes with probability ρ
right
and in this case we have right just
three parameters which
uh define our task that's difficult
we can also think about switching
bandits with
varying difficulty uh which then just
means that
with probability ρ uh the reward
probabilities associated with
each arm either remain fixed on the next
trial so they are just kind of
translated
or they are sampled uh sorry with some
probability
one minus ρ should be here right they
are staying fixed or with probability
 ρ
they are just sampled from a uniform
distribution in this case
beta distribution with parameters one
uh and in this
in this variant of the task right we we
just have
 α and β as a fixed uh difficult as
difficulty parameters
so active one kind of disappears so
you're kind of averaging over ϵ in
the task
um and just to give you an example also
we will not discuss this but how
restless bandits

uh setup looks like is basically you can
uh
uh you assume for example that the logic
transform of
of reward probability just follows a
random walk so it's just a
brownian exploration in the logic space
of the reward probabilities
and this would also require potentially
changing the generative model which i
will introduce but
it's not necessary so one gets pretty
similar solutions
and behavior so um
now uh to come back to
the example from before uh if we have
like
multi-arm bandit with four arms
uh the setup is exactly the same
and in addition we assume that the agent
has access to the underlying probability
of change so this is not something which
is unknown
this simplifies uh the
learning rules and belief update
equations
uh however one can extend this what i
will introduce today to
the setups where the probability of
change has to be
learned also or that the probability of
change is also something which changes
over time so that
right you can have to track how often uh
reward probabilities change in different
times
uh so this will be kind of example of
decision making in volatile right

environments uh
so uh to visualize the algorithm
and the basically the only difference is
practically
that now you have an effective
forgetting
of what you uh you know what what agent
learned before
and one can see this for example if you
look uh
uh on this square to the left
uh as kind of time evolves and agent is
selecting other arms
uh this value which uh
reward probability associated with the
leftmost arm
just decays back to uniform probability
right so in a way
agent is forgetting information or
expectations it had about
this arm and it assumes that with time
the reward probability beliefs about
reward probability will revert back to a
uniform distribution
and the algorithm is really a
straightforward
extension of what i already described
before
so the generator model uh now is
slightly more complex so be
besides the the likelihood
term so observation likelihood which
remains the same is just the bernoulli
distribution
now we have a kind of state transition
term
uh which tells us how reward
probabilities change over time so and

what this means is that if uh
one believes that there is a change uh
reward probability will be independent
from the previous values
and they are just drawn they just
belong to a uniform distribution so this
uh kind of prior
belief and if there is no change uh
our the transition corresponds to a
delta function which just means that
uh uh that the reward probabilities stay
unchanged from trial t to t minus one to
 t
and finally uh like the prior on
each trial we have the same prior about
probability of changes and this is just
again a
bernoulli distribution with probability
 ρ
which we here we will just assume this
is a known parameter
to the agents right so uh
the problem here in like dynamic cases
uh you can still apply
the the bias rule and you can compute
the posterior
both for the change probability terms of
 j_t
and for the post marginal posterior
about reward probabilities however
as you see here the exact kind of form
of the posterior is
not anymore doesn't belong to like
conjugate
so the prior is not any more conjugate
probability distribution to the
likelihood
and this is not anymore a simple beta

distribution but it's
becomes a mixture of beta distributions
and as you are kind of evolving into the
future this becomes larger and larger
mixture of beta distributions which is
uh well practically intractable right if
you kind of expand this to
open any and the number of trials
so because of that we want to have
something which is a bit more efficient
algorithm
uh we can basically introduce a mean
field approximation
when we now say that okay our
probability distribution can be
described as a product of a bunch of
beta distribution and a categorical
distribution which
just tells us the probability of change
right on trial t
uh and uh
how one what what this corresponds to
here so basically uh
uh what is actually we are using here is
a bit of uh it's not a standard
variational inference so where you would
have to kind of compute the gradient
over the radiation free energy to find
the optima this simplifies the things
because you just need one step
to update parameters both about change
and change probability and about right
reward probabilities
uh
this makes it not super optimal so there
are better solutions how i can do this
but it's very efficient so and
in in the end for the bernoulli bandits

this is there is will not be much
difference uh you can use better
algorithms
by uh this gives you just marginal
advantage
on the long run uh just because the
the problem is very noisy and it's very
difficult to actually
figure out the correct choice so what
this
variational smile does so it was
introduced by basilic
uh in well quite recently
2021 uh
so they provide you a bit more detailed
justification for what i'm
what i'm saying here i'm just
paraphrasing paraphrasing a bit
uh how the algorithm is defined so
basically we can
associate the marginal about change
probability with the exact
posterior marginal because you can
compute this analytically
and then we use this as a
basically uh
known known belief about change
probability to estimate uh
to estimate re reward probabilities by
right averaging in the log space uh over
over different
uh prior beliefs about
reward probabilities right so basically
you're instead of
averaging in the probability space
you're you're kind of um averaging in
the in the log space here
uh what this kind of results is very in

a very simple set of update rules so
on the left side we just are just
showing how ω_t
is updated and this is um
just correspond basically to
forming beliefs about change probability
based on bias factor
shown here right which is just the
likelihood between uh
observing o_t given that the change
didn't happen and observing o_t given to
the change happened
right at the current trial
and then using that estimate to update
uh your beliefs about different arm
probabilities
and basically um depending whether you
selected the arm or not so
right basically the ω term here
plays this a forgetting
rate so the uh the larger the ω the
closer to one so the
larger the probability the change occur
the more you will revert back to the
 β_0 and α_0 parameters so the
initial prior belief
unless you will depend on your current
uh current beliefs
from the kind of previous trial uh
and this also has a important limit like
right if the agent believes that there
is no change in the environment you will
you will revert back to the exact
inference and the update rules which we
had for stationary case right so
uh and that's kind of also nice thing
about this
algorithm it can be just generalized to

any

any knowledge about probability a change

probability

okay so uh the action selection

algorithms did not

change so so the learning rules will

change but

we are practically doing still the same

way of making

action selection so for thomson sampling

there's a sampling from the posterior

beliefs

uh for the asian upper confidence bound

it's slightly different so we are

using kind of the mixture between

uh again the mixture of possible

parameter

values to estimate the inverse uh

because from the this predictive

posterior it's difficult to inverse it

it's a mixture of two

beta distributions so this is just kind

of approximation one uh can use for

bayesian ucb and for

action selection uh approximate uh

expected free energy one gets with

uh a very same uh

uh set of equations because basically

raw

row can just so probability of change

drops out

it can be incan being nor easily there

uh okay so now if we

do kind of the same uh comparison first

of uh

exact and approximate after inference

algorithms we see a slightly different

picture

to what we had before
first it seems that as you increase the
number of trials
and you kind of compute this regret rate
for this specific number of trials
the regret rate does not change so in a
way
algorithm converges very fast to a
specific regret rate and
for different and there is a clear kind
of
stable minima independent of number of
trials you're
you're exposing algorithm to uh
similarly again we see like very similar
behavior between
exact active inference algorithm and the
approximate right
and for this example i will just i just
kind of fixed
 λ uh to uh 0.5
so in it just seems to be
reasonably well uh valued parameter for
many uh
many situations which we see here uh
however so
here i'm showing for 10 arms if we
go to 80 arms the picture slightly
changes so it seems that
optimal λ parameter although it
doesn't
change with t it changes with the number
of arms and obviously with all the other
row and epsilon all the other uh
parameters so this makes things a bit
difficult
and basically suggests that it's
important to kind of find

potentially a learning rule like
self-optimizing
way of estimating λ and there
there is some work
actually in active inference literature
which potentially has
solution for that so uh we just have to
test it out at some point
uh so however or we can still use the
same
value this does not uh influence
that much the performance later on so
what we see
when we kind of compare it to uh
approximate active inference with
bayesian ucb and
thomas sampling is that for a range of
uh
right settings uh after either with like
fixed
um uh advantage of the
of the best like reward probability
advantage of the best arm
uh we see that active inference is just
right uh
performs better
already after like 1000 trials or so
compared to the other two algorithms
this is uh uh just for different
change probabilities so for example row
0.05 corresponds to a change every 200
trials
it should be every 100 trials on average
and the last
step here is every 25 trials so this
would be the most difficult right uh
scenario so one sees is that kind of the
more more difficult scenario here's the

the differences uh disappear so you're kind of losing the advantage uh however um if sim okay similarly if we look at the if we fix number of arms for example 240 and we just change the epsilon parameter a similar trend is visible right that uh the more easier the task is the bigger the difference uh in like non-stationary scenario you get relative to to other algorithms of course one would potentially uh kind of destroy this advantage if uh changes become very slow like every 10 000 trials or something where you are kind of approximately in the stationary stationary world right and okay so this would be like this switching bandits with varying difficulty we can do the same kind of analysis for switching bandit uh sorry previously we looked at the switching bands with uh fixed difficulty similar analysis can be done for varying difficulty and uh interestingly one sees here uh notes because epsilon drops out one sees here uh actually for the exact active inference there is like a clear minima which uh seems independent on the kind of any of these parameter terms the so ρ or k so basically we can fix λ to 0.25 for the exact acting difference algorithm and we can still keep for the

approximate

λ to for the approximate active

inference algorithm λ to 0.5

and just to right show what happens

here is that there seems to be like

invariant difficulty there seems to be

also advantage of using

exact active inference decision making

algorithm

uh it outperforms the approx

approximated quite clearly

uh in this settings and it doesn't

require a fine tuning in this sense

right for the range of problems you can

have much better performance with the

single uh

choice of λ value uh

and again so the in this case

interestingly the easier the problem

becomes so

in this kind of quadrant the less the

difference

uh between algorithms is and the

actually bayesian ucb algorithm becomes

quite uh

quite efficient in these settings also

here right you see that

beijing ucb also achieves uh quite good

performance

interestingly much better than

optimistic thompson sampling which

for me at least i didn't found any paper

who

previously uh showed something like this

so yeah

this is also a bit probably new result

for

machine learning people uh

okay so just to conclude right
active inference does not result in
asymptotically efficient decision making
uh
additional work is required to establish
theoretical bound and regret
and derived improved algorithms in
non-stationary bandits however
we see better performance
in comparison to other algorithms and
especially noticeable in more difficult
settings right that
you're kind of uh getting uh
in the more difficult task is you're
kind of forgetting the better results
a tentative to-do list for like next
steps here is like
introduce learning for the λ
parameter establish
a really like kind of theoretical bound
on cumulative regret so what
can one expect to see
given different choices of the algorithm
for action selection
uh based on expected free energy and
so right this would hopefully improve uh
behavior and stationary cost
a case and potentially also apply to
some like
real world examples so this kind of
right uh
in into in machine learning field i
would say right you know
this kind of recommendation systems uh
optimization problems and similar right
just to see how it performs in this kind
of scenarios
yeah yeah okay that

i would just like to thank all the
collaborators on the project
uh and the people who helped me with
different advices
and you can also uh
find the slides here and the code is
available
my github page i would just not
recommend to
use it in the next two weeks because the
paper is under revision and
i'm breaking stuff constantly yeah
awesome thank you you can unshare and
we'll
return to just um discussion for the
last
45 minutes or so here but thanks for
that awesome
presentation with always good to get
multiple um multiple times just to sort
of
see some of those figures in the paper
then also there are some different
figures and some different views so yeah
again anyone in the live chat is welcome
to ask a question
or anyone who's here in this jitsi
i'll ask um a question from the live
chat first and then if you're here in
the jitsi
please feel free to raise your hand so
um it's written in a live chat since the
bandit problem here
is not markovian does this mean that we
only need to consider
the current time to calculate the
expected free energy
uh why why is not the mercurian

well let's let's actually clarify what makes a problem or a situation markovian or non-markovian

okay so uh for example uh the problem is markovian because the changes are only dependent on the last trial or the current trial uh right so the reward probability in this non-stationary case uh will be a function only on the reward probability in the previous time step and this is what uh makes the problem actually markovian so i i didn't talk about there are non-markovians bandits but this is not what we are discussing here right it is really just markovian but this is not the case why you don't need

well maybe yes actually i mean if you would go to non-markovian bend it then you would potentially need to plan ahead for longer because that would require that there are kind of dependencies between your actions and outcomes which you observe in a sequence and different sequences might result in very different outcomes so right because here we are in markovian case and uh agent cannot change the dynamics of the environment in any way so it's in a way just kind of passive uh sampler from the environment uh then you just there is no gain in planning ahead so

you can just re-evaluate your beliefs on
each trial
so it's kind of like a memoryless
process which you
brought up and then what would have to
change to maybe account
for situations where sequence of actions
does matter
uh so for example one could
introduce structural dependencies
between different arms
so that for example rewards uh
reward probabilities depends on the
location
and allowing agent only to
to select uh to make their choices from
the nearby uh
right kind of introducing spatial
dependency
that would be one example where then
depending on in which
part of the space i am in kind of where
i selected
one arm this limits me to what what's
the next arm which i can select and this
would require you to then plan
depending on uh where you should be in
different trials depending on how you
expect
things to change right
all right this this would kind of
introduce then uh
the requirement for planning
cool very interesting um
blue or dave or sarah wanna ask anything
otherwise i
have some questions
you mentioned a few industrial

applications

and a few ways in which people do use

the multi-armed

bandit this is just kind of like um a

logistical

question like what is the rate limiting

step in

those use cases deploying active

inference agents

is there a way to kind of wrap the

inputs and outputs of multi-armed

bandits

uh in a way that's sort of interoperable

like you talked about how the learning

rules were similar but then you

you juxtapose different action selection

approaches

so in the context of pipelines that

people are already using

is there a way to kind of hot swap

active inference

and maybe have it be deployed in

industrial settings very rapidly

well yeah i mean that would be one idea

i mean for example uh

if you have these non-stationary

problems and you have already been used

using thomson sampling for example or

even like optimistic thompson sinclair

as a choice of the algorithm in a way

you already have a way to

form beliefs about relevant aspects of

the

of the problem then you can like really

easily swap

right the two and you just can then

apply

the the difficulty just to figure out

how to compute basically expected free energy
and test it out right for the different generative model
that people might probably have
yeah uh again for stationary problems that's maybe trickier
it might work in some situations but
yeah
you you don't have this kind of nice asymptotic guarantees that you will always find good solutions
in a way it's an advantage of active inference in a changing world like people's preferences for a given advertisement or the the situation for trading is always changing and so it's a false allure to have something that has extremely well behaved behavior in the asymptotic or infinite case because we're not in the infinite case we're in the finite and dynamic case
and so it's almost like the sort of strong pillar that purportedly is underpinning these other approaches isn't so much of a gain pragmatically so it's pretty cool to hear about that
blue thanks for the raised hand what would you like to ask
so i have a question that was left over from the dot zero video and something you kind of alluded today
in your talk can you kind of detail the difference between the switching bandits

and the restless bandits i was like
unclear on like the timing of the
switching
in the switching bandits and then also
um just like a part b of that question
in the case of the restless bandits um
what are
are the similar like algorithms that are
optimal for switching bandits
also optimal for restless bandits you
had mentioned
the active inference was good but what
about the others that are commonly used
no it's the same right i mean
so
let's say in restless bandits the only
kind of
so in switching bandits we have like
piecewise linear problem or piecewise
stationary problem
which means that between trial one
and trial ten everything behaves
stationary and then when the change
comes
you're just getting uh new reward
probabilities associated with each other
right
for example that would be a kind of idea
of a switching bandit so that
between changes everything is like fixed
and the restless
bandit assumes that things continuously
change and
here the example i gave was one can
assume for example that
reward probabilities can be described as
a
random walk in this

in this logic space of the probability
so basically because probability is like
between zero and one
you transform into one constraint space
between minus infinity and infinity
you just have a then gaussian random
variable so to say
and then you back map it back into the
into the with the sigmoid function for
example into the probability space
uh and and uh so but so for example the
algorithm i show like for belief
updating you
one could also just apply it there so it
just one one doesn't know what would be
the row
so what is the change probability in the
restless case
so in that sense you would need an
algorithm which can also infer the
change probability
and the restless case doesn't
necessarily translates
to kind of potentially fixed change
probability
uh so it's a bit more
different problem so because the maximum
arm the changes between
right between arms which are optimal uh
do not follow specifically like
the same structure as in the switching
case so
uh one can either take a different
generating model which actually assumes
explicitly the random walk and like for
example
uh hierarchical duration filter is
something which one could apply there

but they're also uh like what other
belief updating
algorithms for that so the local or the
global
maxima and minima are changing in the
restless as well
yes yes yes so this would be kind of
this case of uh
varying difficulties over the relative
uh
probability between the best arm and the
second best arm
varies all the time
yeah i don't have
i could i could have just drawn the
lines
to show but i don't know i didn't do
that i i could hear that
i just don't have it right now something
to
illustrate this so one other thought on
the advantage possibly of active
inference is that with a deep
generative model using the same skeleton
of maybe even the same code it could be
possible to
do model testing between two different
types of bandits like what kind of
scenario am i in
or even have deep temporal models so
it could be extended in a way where a
sort of instantaneous sampler
might be led astray yes
yes yes i mean in principle you can um
any hierarchical asian generative model
which consists of multiple models
potentially you could
also write generalized to thomson

sampling

or by asian ucb i just wouldn't assume
that this would be very efficient
way to figure out which is the correct
model

which you should be currently applying
to the specific task and this is
potentially also where active inferences
uh would provide an advantage in
such

scenarios right where you can also kind
of

learn about the generating model itself
better over time

and that made me wonder what would it
look like at the sort of human level as
we're making decisions that are
sequential

in our day our decision making what
would it

look like or what would we keep in mind
if we were going to be making decisions
more like an active inference agent than
like a thompson sampling agent
like what would be you know when you're

in the grocery store looking at the
cereals that you've had before or not
how would an active inference agent
behave differently um

just kind of wondering yeah i mean
that's a good question i mean we could
actually we have some data sets
which which we could use exactly
to ask this kind of questions yeah i
mean i don't have on top of my head any
clear answer um

what what would be your expectation sir
well i guess um it would tell you one is

a

or better tell you when is a good time

to try a different cereal

because you may like it even more

especially if cereal

recipes change over time

and yeah then you have some likelihood

to be stuck maybe

with the super bad cereal but also i

like you to

have the one you like most

yeah i mean i guess this exploration

wouldn't would be more structured in a

way more

directed right um

right like i mean when the ingredients

change you check the ingredients and

then now you've updated your likelihood

of

trying something new yeah all right i

mean

thompson sampling doesn't have directed

part of the exploration

just the random exploration

whereas they're here the focus would be

more and right directed

exploration and potentially

thinking how to add random

so yeah one other piece is like

we're often comparing and contrasting

active inference to

reward learning and reinforcement

learning so it's almost like

instead of making that decision based

upon the highest expected value

like choosing something proportional to

its relative value or always going with

the one with the best likelihood

of having the best tasting cereal there
can be some
other heuristic and so it opens the door
to just purely curiosity driven
sampling like i've just never had this
and it's not
even as much a reward maximizing
maneuver as it is just a purely
epistemic
gaining maneuver and then as we've seen
when there's a pragmatic and an
epistemic component
to the function that's being optimized
then
those decisions can kind of co-exist and
be put on a common grounding
unlike in a purely value driven
framework
where even the exploration has to be
kind of
coerced back into reward
yes mean so i think uh syed noor and she
had like as a first author she had an
interesting paper the mystify
active inference and they discuss a bit
about
learning the preferences themselves
right and what kind of consequence this
has and this kind of
puts a different perspective on
understanding what the reward is because
in real world you don't necessarily know
why should something be more rewarding
than something else but you learn
this over time and you just learn to
prefer
different outcomes they don't even
necessarily have to be

in somehow rewarding more in the
absolute sense
you just build experience with some
outcomes over others and you'll
start to prefer them and this
then appears as if you would doing
reward based
decision making but i mean it's actually
in a way preference based decision
making right
cool i i had another question about the
approximation of active inference
is that the only way to approximate
active inference are there
other moves that you could have made to
approximate
the sections that you did approximate
are there other pieces that can be
approximated
what can be swapped or approximated
but still retain this essential
structure of an active inference model
[Laughter]
um
well i don't have many ideas what else
one could do
i mean the problem is in this scenario
it's relatively simple
right but for example one can kind of
think of okay
given that we have a
to compute the expected information gain
and we are kind of learning of having a
way to
to estimate it efficiently like just
with this approximation
uh there is other way how you can
compute this just by sampling

and you can draw a couple of samples
from your beliefs and you get some
estimate on the expected information
right so this this would still be in the
kind of
active inference frame framework
but just like a different way of how you
actually compute expectations and what
they will
kind of represent there right you will
this would be a way to add a bit of
random randomness to the decision-making
process
i mean active inferences also general
enough to work with many kinds of
approximations right and you can
do approximations and many different
points but
as long as you they get some sort of
variation of free energy
i guess it's still within this framework
and
immediately separated the
inference part from the action selection
part but i think
actually you can use it as a model of
both
right and then you can do approximations
and different points in this joint model
and as long as you can still write down
a variation every energy
i guess it's a counter active inference
well i mean i don't know even if that's
necessary i mean i would say
any kind of bayesian belief updating
even if it's not specifically motivated
by variational inference
would it the consequence have minimizing

variational free energy once once you
compute it from some posterior which you
obtain
and i mean right it's it will still be
in a way
active inference so right in a way one
can replace
like smile variational updating with any
belief updating rule
and still keep the same
concept there so because we are just
getting a better bound
on the marginal likelihood if you have a
better posterior
so so what would you say you know anyone
is necessary or sufficient for
a model to be considered active
inference versus not active inference
just including sense and action or
perception
inference action or agents in a niche
or blanket states these things are
sort of we're in an overlap of venn
diagram with certainly many
classes of models different approaches
and is it like that blurry intersection
that we're looking at and that's where
action and inference
are being just applied together or is
there something unique or something that
we can use as a diagnostic
well i guess the difference would be
more than this kind of uh
this action selection i mean planning is
inferencing part not then
not necessarily as perception in its
inference because as i said
one can think of many ways how one can

solve that part
but kind of once you go into this
planning as inference part and
also concept that right your actions
themselves
will change beliefs and you will
choose actions which are the best in
changing your beliefs or what you want
to achieve
then this is like i mean idea of active
inference in a way it's uh
let's say it's circular
inference problem right
uh it's not in a way it's any
more so easy to disconnect in the
effects of choices and effective
perception
they all depend on each other and also
being aware of your
own uncertainties and i think
if you do standard reinforcement
learning where you just
calculate expected rewards i don't
think you're always aware of how certain
or uncertain you actually
are about your own generative model for
example
and then potentially with active
inference
it's easier to know which action to take
in order to better learn your model
very interesting it's almost like by
carrying and propagating our uncertainty
and having a self model of action and
learning
then we get that almost like second
order
cybernetics where we're acting

in a way where in the future we can
expect to learn better or expect to act
better
as opposed to just this hungry search
for the best action and then learning is
only a one-step
projection into like what action is
gonna be informative
right now uh what's the next wikipedia
article that's most informative
rather than what's like the trajectory
that i expect is going to be resulting
in more effective learning
or action again on a common grounding
so that's pretty interesting looks like
you have a thought though
dmitry no i mean uh
the yeah that that's i mean definitely
important part but i mean i would then
call thompson sampling would also
satisfy that beijing ucb right i mean
they all kind of
are any bayesian decision making
algorithm
would necessarily have to take into
account uncertainty
uh
if it's derived from a bayesian
decision-making theory but
yeah it is not that clear that by this
aspect that
okay the actions also have a consequence
of reducing your uncertainty in the
future and this is what you can use also
for
that's a gauge on which one which action
is better
so yeah um

subtle differences and not that obvious
always
it's it's one reason why we're so
interested in
in ontology and slowly scaffolding the
research so that we can actually
juxtapose different models and
understand
where they differ like okay they're kind
of it's like two road trips and then
this person just took a little extra
loop or
this bridge they crossed this way versus
this person you know
took a different route right here we can
understand like
you talked about the smile variational
updating but then just recently you
mentioned that there's
that's sort of like a module you can
switch out
so what is smile or
because i also noticed a very recent
citation
so what does it do or what is it
different in regards to other ways you
could have fit that module
in
i mean i have implicitly used smile for
years now it's just a very simple way of
updating your beliefs so uh
so the problem a bit with kind of uh
variational inference if you want to
find the
minima of approximate posterior you kind
of
necessarily need to iterate uh
through several loops like minimizing

the gradient uh
so following the gradient of the
variation of free energy right
and i mean this also doesn't make it
very efficient
uh in for
this kind of applications so for me at
least right this variation smile
approach
side steps this iteration so you can
kind of transform variational inference
into just a single update step
uh just because there is a way part
which you can
compute explicitly and you just assume
okay this is now my fixed belief about
that
and the other part which you need still
to update
through variational approximation
[Music]
uh and so this is i guess
a small advantage here
but as i said right there are other
approaches how you can com
update beliefs in these scenarios and uh
you know in the bayesian sense there can
also be more
uh optimal but they just bring you
closer to the exact posterior
so but i mean we tested these things
uh with some examples and
the story remains the same so there is
no gain on
making things more complex on that on
that
but one can also imagine that like in
different

environments better generality models
and approximation rules
would take you further so but for
bernoulli bandit this is just
not the case what kind of empirical
data sets are almost
whether they're open source or just
obtainable are amenable
to this kind of analysis like if
somebody hears about the algorithm
and they kind of want to see it in
action or play with it themselves once
you're
you know two weeks from the recording of
this when your code is available like
is it possible is your code set up for
more of a simulation or is it something
where
we can plug in a type of data set that
might already be structured
appropriately
uh it's currently just for simulations
but i mean one could potentially write
play out with different algorithms add
other algorithms
both under like a learning part and the
action selection
i mean that that would be super useful
for everybody
yeah if there are people interested in
that where that would be
our repository is open and welcoming any
contribution
it was part of the structuring of your
paper that
like made us excited to juxtapose
it with these other approaches is you
kind of showed that you can

directly compare active inference
alongside other models
obviously it's something we're coming
back to because that's the crux of the
results of the paper
is really the different dynamics as
time increased or as the relative
challenge increased as the number of
arms was changing
between these different styles uh
and the two styles of bandit so it's
kind of a cool thing that people could
both
build on directly what you're working on
but also
maybe more broadly instead of just a
single model being presented in a paper
people could just include multiple
types of models in their sort of
baseline paper
so that we wouldn't just have to read
the paper that says this algorithm works
well
we could see them directly compared
and that's something that more and more
papers are doing with active inference
yeah i mean the
the kind of also
the libraries we use is also this is
something
which was developed relatively recently
like
uh by google jacks it's kind of uh
acceleration for linear algebra algebra
libraries so which allows you to run
very fast
this code and it also integrates well
with the probabilistic programming

language so it's a
numpyro which was also uh
developed really recently so one could
be in practice
link this to any behavioral data or to
create some test data
uh very easily uh
with a few lines of code so yeah there
is also this perspective
what language is the code in one or
multiple
and then what areas of conceptual math
do you think somebody would want to know
before diving in
so the language isn't i mean like
program language is python and just
depends on as i said couple of libraries
like
jacks numpy very standard stuff except
the jacks that's that's
new so yeah somebody
interested in using it would have to
learn a bit about jacks but that's
a long term also useful so go for it
but from the matte side yeah i'm
well i mean maybe getting a couple of
introductions to multi-arm bandits that
would be kind of
a good place to start uh right there
there is a
there is a recent uh
quite recent introductory paper mostly
on blended which exposes
many different algorithms in the
stationary concept so uh
provides a bit insights about this uh
historically
historic way how they analyze uh these

problems

so yeah uh i would suggest that that's

the kind of potential

place to start cool blue

so i'm curious about this google docs i

haven't heard of heard about it but

like i thought google had their own

language like don't they have like

golang right isn't that google

or tensorflow right

well yeah tensorflow

yes they have tensorflow and i think the

basis is

uh similar xla

uh it's called like accel accelerated

linear algebra

uh for both tensorflow and jx

so what jax is is basically accelerated

numpy

so you can just run standard nupi

code and in pure python more or less

uh and get very um

so you you get for free like uh um

computational gradients so it's kind of

autograd

library so you can kind of the compute

gradients on very complex graphs

uh this is also something which

tensorflow allows you

but this seems to provide more more

benefits for this dynamical scenarios

so i have problems when i tried learning

or using tensorflow it's very difficult

to think

into right code which is dynamic there

and that's why i i never actually start

started using it i started with pytorch

at some point

and now that jax showed up that has some speed up advantages it's also kind of quite lucrative

so one code question just to stay on this theme and then ask a question from the chat

what about the um python implementation like uh i think infer actively or what alec chants at all have worked on what are the similarities or differences with their python approach there

i think currently none because i'm also involved in that so they will also start using text okay i guess

cool so that those threads will join together

okay uh but i mean this also code just like produces some of the

i mean one could also write use just the spm code but this is super complex uh right it would be super slow so for example just to examine numerically this stationary scenario this would take months probably too

[Music]

okay nice somebody's ranking

uh so yeah um

right that's uh uh that's kind of the the problem there and this i wrote the code to be kind of very efficient in a way so just kind of removes lots of complexity that you don't need

cool uh all the right in some scenarios you want to have general

uh description of the program so yeah
so here's a question from stephen in the
chat

do you think that parallel modeling
processes might be used more in the
future

with different model approaches
highlighting different patterns of
behavior

happening in different niche contexts
i would say in some ways it's what your
paper did but what are you thinking
about yeah

well i mean from practical side yes i
mean that's what

you should be doing because just figure
out what works best

and just don't don't think

too much about well the philosophical
side of things

uh but yeah but i mean i guess long term
one could i mean imagine scenario where
more and more things become
generalizable to this description

so it's just in a way a
difficult process takes time even for
those who are

just learning about the technical
details like the visual

tell for me was that the figures were a
grid

of graphs so it was kind of like
three different settings of difficulty
or three

or five different settings of arms it
wasn't just showing

we ran it with one parameter combination
each of those graphs was a grid of

combinatorics and then you presented
what i guess could be considered two
different niches
with the static and the dynamic and then
today in the presentation we
heard about all these other variations
and so those are like
kind of toggles in the code you can say
i'll take
no a1 alpha or i'll take
a2 beta as far as the combinations of
how to run it
and so as the code becomes more
interoperable
and pruned down to really the necessary
pieces
then it becomes easier and easier to
expand it back out
so that we can choose amongst different
options
for a given piece and so that's like
this skeleton
that so many variations flow off of
yeah um
cool well in our last few minutes here
where would you like to uh take it
next week or beyond i mean what are your
current interests or curiosities
well i mean for me it's just important
to have this paper out so next time when
a reviewer asked me
what about that and that algorithm i can
say okay look
we analyze this it's like that it's
similar or different
you can you can get whatever you want
basically there
uh right i mean because you have these

adapted parameters

in a way you can generate very different behavior one can also think okay if you have like an agent which behaves as a thompson sampling what would be the corresponding parameter in active inference framework which would emulate that can you actually even differentiate between them

so uh i mean this is uh potentially interesting questions which one can try to answer and which are just from

my work kind of relevant because of this constant uh questions i'm getting in during the review process

uh and that was actually yeah but then there is also this like quite interesting side in like machine learning

uh where this can find potentially quite interesting applications

uh i i got started a bit uh recently working with kind of monte carlo research

and interestingly uh sorry this is something which was applied in active inference as a way to compute expected free energy in complex problem but turns out that

if you have kind of very complex problems you can also use thomson sampling in monte carlo research

uh right so as a way to just figure out what is the best path to follow uh in a sample

and this potentially provides if this stationary scenario can be improved somehow and uh this is also kind of you can apply active inference to planning inside active inference itself you know in a kind of uh hierarchical circular way though right so what does that look like or how does it get implemented or how is it different than just straightforward active inference uh so the problem is like that when you have quite a complex decision making in a planning problem where you have multiple branches in the future which you have to go it's like not practical to compute everything uh so what people do is a very popular way to do is like monte carlo research where you just sample different paths you you estimate us on a sub sample of possible paths what is like uh the best path to to go and this would correspond in active inference like you're estimating expected free energy of a path in the future uh just through a sample but then you have a problem okay how do i select the same the paths in the sample and then you can apply it active inference to the path selection itself so right you can kind of choose which paths i should sample randomly

when i'm trying to approximate
uh expected free energy for this problem
itself
so that's kind of what makes it kind of
interesting is a possibility to explore
and the way that you just framed it
as well as what we've seen which is the
relative
strength of active inference in dynamic
settings
it's consistent with a lot of the
qualitative and philosophical ways that
people
are talking about active inference as
like a sense making
or a way finding or a navigation
approach
rather than a sort of cut and dry
calculus of decision making just
resulting in
the total you know crystal path just
being laid out before you
it's really about the instantaneous
actions that we take now
in light of uncertainty about the
present and really the past and the
future as well
so it's always cool to see how the
technical developments while they're
like kind of weaving and recombining
they proliferate and then we see oh
actually these three are kind of
interchangeable or these three are
complementary
and then we get more technical detail
and speed ups while we also get more and
more
clarity on what the structure of this

sense making problem is
yeah i agree
any final thoughts here from anyone
otherwise this was a super
interesting presentation and questions
so we really
appreciate it
thank you cool yeah thank you
yeah we're looking forward to for the
next week great
so thanks everyone for joining and
everyone's welcome
to join live for next week when we'll
continue the discussion
and the dot 2 is kind of like our
jumping off
into the the unknown unknown instead of
just the known unknown
so thanks again for joining and we'll
see you next week
thanks bye bye